# The ternary module of the residual Steiner graph on 176 vertices

Private derivation and finite certificate: independent review pending

This note supplies the remaining module and full automorphism assertions for L050, Theorem 1(c) and Proposition 4: Leemans–Rodrigues, *Linear codes with complementary duals from some strongly regular subgraphs of the McLaughlin graph*, Math. Commun. 21 (2016), 239–249. We use the source's residual Steiner construction, the shared incidence calculation, and the separately completed exact weight enumeration. The new finite certificate uses the same matrix-algebra and minimum-support arguments as the companion treatment of the 120-coordinate modules. It does not enumerate the $3^{21}$ codewords again. Statements identifying *all* irreducible representations remain separate retained inputs.

## 1 The common incidence calculation

Let $\mathcal{D}$ be the fixed $S(4,7,23)$ from the shared construction, and delete a named point $\infty$. Let $\mathcal{H}$ be the 176 heptads avoiding it. The graph $\Gamma$ on $\mathcal{H}$ joins two distinct blocks whose intersection has size three. This is the complement of the intersection-one graph in the source's Section 4. The construction checks the Steiner axiom and the intersection sizes, so agreement is through the actual blocks, not just the four strongly regular parameters.

Let $H$ be the incidence matrix with the 22 remaining points as rows and $\mathcal{H}$ as columns. All code spaces below are over $F = \mathbb{F}_3$. The parameters of the Steiner system give $77, 21, 5$ blocks through a point, pair and triple, respectively. Consequently a remaining point lies in $77 - 21 = 56$ residual blocks and two lie in $21 - 5 = 16$. Thus, over the integers,

$$HH^{\mathsf{T}} = 40I_{22} + 16J_{22}.$$

The adjacency matrix $A$ satisfies, over $F$,

$$A = H^{\mathsf{T}}(J_{22} - I_{22})H, \qquad HH^{\mathsf{T}} = I_{22} + J_{22}. \tag{1}$$

For the first identity the diagonal entry is $49 - 7 = 42$, and off the diagonal it is $49 - |X \cap Y|$, which is zero modulo three for intersection one and one for intersection three. This is the common incidence factorization already used for the residual graphs on 120 and 253 vertices; its coefficients are displayed here to fix the precise instance.

Put $V = \{x \in F^{22} : \sum x_i = 0\}$ and $C = \operatorname{im} A$. Since $I + J$ is nonsingular, $H$ has full row rank. Also $J - I$ has kernel $Fj$ and image $V$, where $j$ is the constant vector on 22 points. Hence

$$C = H^{\mathsf{T}}V, \qquad \dim C = 21. \tag{2}$$

For $x, y \in V$, their image scalar product is $x^{\mathsf{T}}(I + J)y = x^{\mathsf{T}}y$. The space $V$ is nondegenerate because $22 \neq 0$ in $F$, so $C$ is LCD. The constant vector on 176 coordinates is in $C^{\perp}$: its pullback under $H$ is $56j$, orthogonal to $V$.

For completeness, write $B = J_{176} - I_{176} - A$ for the complementary adjacency matrix. The parameters $(176, 105, 68, 54)$ give $A^2 = 2A$ over $F$, and $AJ = 0$. Therefore $AB = 0$. Since $A + B = J - I$ is nonsingular ($176 - 1 \neq 0$ in $F$), their images sum to the whole space. It follows that $\operatorname{im} B = \ker A = C^{\perp}$ and $\operatorname{im} A = \ker B$. Together with (2), this proves the two complement–dual identities and $F^{176} = C \oplus C^{\perp}$ asserted in the source.

## 2 Two general certificate arguments

We recall explicitly the two short arguments used for the earlier 120-coordinate modules. They apply to a code in any length and do not depend on the special group name.

**Lemma 1.** *Let $G$ act linearly on a $d$-dimensional $F$-space. If $d^2$ words in its generator matrices are linearly independent in $M_d(F)$, the module is absolutely irreducible.*

*Proof.* Those words span $M_d(F)$, since that is its dimension. A $G$-invariant subspace is stable under every word and linear combination, hence under every endomorphism. If it contains a nonzero vector, an endomorphism can send that vector to any prescribed vector, so the subspace is the whole space. Linear independence of the same matrix words persists under every field extension, proving the absolute assertion. □

**Lemma 2.** *Let $C \leq F^n$, and suppose its complete set $\mathcal{S}$ of minimum nonzero word supports is known. Form the bipartite incidence graph with coordinate vertices and one vertex for each distinct support, giving the two parts distinct initial colours. Repeatedly refine each colour by its old value and the multiset of neighbour colours. If successive individualizations of coordinates have refined cell sizes $m_1, \ldots, m_r$, and afterward all coordinates have different colours, then $|\operatorname{Aut}(C)| \leq \prod_i m_i$. Here automorphisms mean coordinate permutations, as in L050.*

*Proof.* Every code automorphism preserves weight and permutes $\mathcal{S}$, so it acts on this coloured incidence graph. Every refinement is preserved by automorphisms fixing the previously individualized vertices. The orbit of the next chosen coordinate is therefore contained in its current colour cell. The orbit–stabilizer formula bounds each successive stabilizer index by $m_i$. At the end all coordinates are fixed. The support vertices have distinct coordinate neighbourhoods, so they too are fixed. Multiplying the indices gives the stated upper bound. No claim that a colour cell is an orbit, or that every support automorphism preserves the code, is needed. □

## 3 The exact module and group certificate

The literal permutations in `module_data.json` preserve every adjacency entry of $\Gamma$. Their closure, computed by exact permutation composition, has order 443520. They also permute the 22 point-incidence rows of $H$. The resulting faithful degree-22 action is explicitly identified with `MathieuGroup(22)` by an isomorphism checked in GAP. Thus the subgroup $G$ they generate is $M_{22}$. This conventional group identification is distinct from proving that $G$ is the *full* code automorphism group. The latter will follow from Lemma 2. The auxiliary group calculation also verifies that the degree-176 action is primitive with stabilizer $A_7$ of order 2520.

The row-reduced adjacency basis is checked byte for byte against the basis of the retained complete enumeration. For each coordinate generator, its images on these 21 rows give a $21 \times 21$ representation matrix. The checker reconstructs those images from the proposed matrices, so the matrix module is the actual code $C$, not a representation with only the same dimension. Deterministic multiplication and elimination produce 441 independent generator words, all of length at most eight. Each word is evaluated again and the 441 flattened matrices are row-reduced. Their rank is 441. A separate GAP calculation confirms the same generated algebra dimension. Lemma 1 proves that $C$ is absolutely irreducible. This action is faithful as well: the 176 columns of a code basis are distinct, so a coordinate permutation fixing every codeword fixes every coordinate. Column distinctness is checked in the certificate.

The retained complete enumeration gives $d(C) = 56$ and exactly 924 words of weight 56. The supplied literal weight-56 word belongs to $C$. Its orbit under $G$ has 462 words and is disjoint from the orbit of its negative. Together those two orbits have 924 words, so they exhaust the minimum words. Their supports are the same 462 distinct subsets. These assertions are checked by finite orbits and the already accepted count; no new exhaustive codeword computation is involved.

For this support-incidence graph, with $176 + 462$ vertices, the deterministic refinement rule of Lemma 2 successively individualizes coordinates $1, 5, 29, 61, 2$. The corresponding cell sizes are

$$176, \quad 70, \quad 6, \quad 2, \quad 3.$$

All coordinate colours then differ. Therefore

$$|\operatorname{Aut}(C)| \le 176 \cdot 70 \cdot 6 \cdot 2 \cdot 3 = 443520.$$

Every graph automorphism preserves its adjacency row space, so $G \le \operatorname{Aut}(\Gamma) \le \operatorname{Aut}(C)$. The already checked order of $G$ forces equality throughout:

$$\operatorname{Aut}(C) = \operatorname{Aut}(\Gamma) = G \cong M_{22}. \tag{3}$$

This upper bound uses no graph-isomorphism backend. Such software was used only to discover the two literal generators.

The complete retained weight distribution also determines the dual one by the classical MacWilliams identity

$$A_j^{\perp} = 3^{-21} \sum_{i=0}^{176} A_i \sum_{s} (-1)^s 2^{j-s} \binom{i}{s} \binom{176-i}{j-s}.$$

The certificate computes all coefficients exactly, checks nonnegative integrality and total $3^{155}$, and obtains zero at weights one through five and 184800 at weight six. Thus the parameters are $[176, 21, 56]_3$ and $[176, 155, 6]_3$, with the asserted minimum counts.

## 4 The precise source boundary

Theorem 1(c) describes the *unique* 21-*dimensional* irreducible $F_3$-module for $M_{22}$. Proposition 4's final sentence omits the dimension, but its proof immediately restores it and invokes the known 3-modular character table. We retain that intended classification statement, and the proof's further assertion that 21 is the smallest nontrivial irreducible dimension, from the source's reference [13]: Jansen–Lux–Parker–Wilson, *An Atlas of Brauer Characters* (1995). These assertions are understood up to module isomorphism. Our full matrix-algebra calculation proves irreducibility of this particular module; it does not classify all modules. The unqualified literal uniqueness sentence cannot be correct, since the trivial one-dimensional module is irreducible too.

The source's historical near-optimality comparison for the dual is a separate assertion based on its tables, not a current table claim or a consequence of (3). No additional irreducibility is asserted for the 155-dimensional dual, which contains the constant line. The group in (3) consists of coordinate permutations; allowing scalar monomial maps would change the automorphism convention.

The complete mandatory local files are the shared Steiner constructor, the literal data, the certificate and the GAP identification, together with the retained exact enumeration data and its separately reviewed method. The orchestration driver and its weakening controls are listed separately. No Lean formalization or whole-L050 admission is claimed in this bounded supplement, and no software-inclusive compression figure is inferred from it.